pfi

Python for Fixed Income

COURSE 8 · IMPLEMENTING AGENTIC AI

An Orchestrator and Specialists

WEEK 3 · DAY 1

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Today: an orchestrator and specialists

Week 3: many agents, kept honest

Day 11	an orchestrator and specialists
Day 12	memory and state
Day 13	evals
Day 14	guardrails and cost control

Today

1. Why split: what every request carries
2. specialist and route_table
3. Handoffs against agents as tools
4. The allow-list, asserted
5. route_table, and the desk brief
6. Parallel tool requests, once
7. What the orchestrator costs
8. Excel SUMIF, Git, the AI check

Every model response in the notebook is a saved response, replayed. A live run can differ.

The rules for agents

On the synthetic book: numbers from tools, described

Allowed names

DV01 per 100 of face

position DV01 in dollars

the largest DTS contributions in the file

Never called

cheap, rich, or attractive

a trade idea or a recommendation

overweight or underweight

On every agent's output: text, a saved run, a brief

Allowed names

the agent returned

the saved run

the brief states

Never called

the agent's opinion or view

what the agent believes

the agent's advice

Why split: what every request carries

```
# characters of tool schemas each agent sends with every request
macro                    2,974 characters of tool schemas
credit analyst           1,481 characters of tool schemas
risk                     6,749 characters of tool schemas
PM                           2 characters of tool schemas
one agent, every tool   11,204 characters of tool schemas
```

Shorter requests

only its own tools

Fewer wrong tools

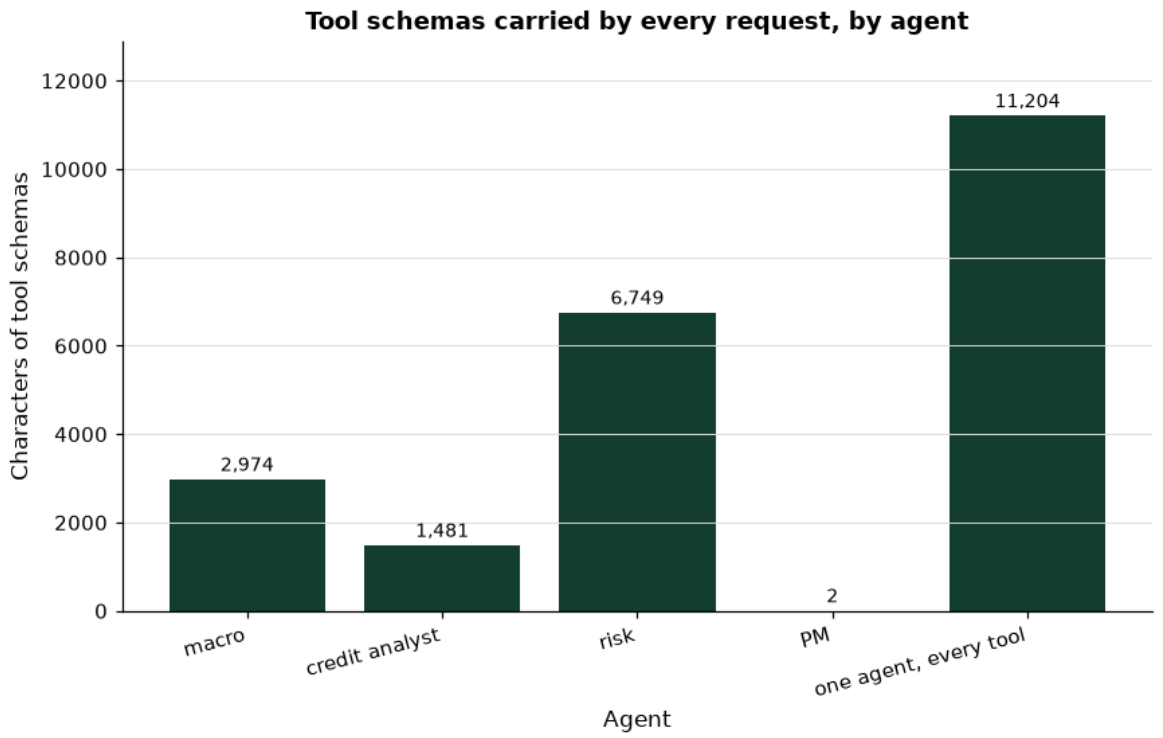
less to choose among

One job each

instructions for one job

20 tools in one agent; the covenant tool maps 11 sections of Article 4 for the credit analyst.

What each agent carries



One agent with every tool sends 11,204 characters of tool schemas on every request; the PM's empty list is 2.

Four specialists, each from an allow-list

```
risk = agentkit.specialist("risk", RISK, RISK_INSTRUCTIONS, aclient, model, 11).clone(
    output_type=agentkit.DeskAnswer, handoff_description=DESCRIPTIONS["risk"])
```

```
macro           tools  6; answer DeskAnswer
credit analyst  tools  3; answer CreditAnswer
risk            tools 11; answer DeskAnswer
PM              tools  0; answer text
```

route_table(result): one row per step: the agent, and a tool request, result, handoff, or answer.

specialist raises when the tools differ from the list. 54 passed with the day's two tests.

Handoffs against agents as tools

A handoff passes the conversation

transfer_to_<name>: one tool per specialist

the specialist reads the whole conversation

its answer ends the run

the orchestrator does not speak again

An agent as a tool keeps control

one tool per specialist, an input it writes

the specialist starts a run of its own

its answer comes back as a tool result

the orchestrator asks several, then answers

```
position DV01 of the book: 306,996.49 dollars per bp; the 10 Yr on 2026-10-02: 5.28 percent
```

The question needs two specialists: the book's DV01 (risk) and the 10 Yr in the file (macro).

Every check passes; the yield is wrong

4.0

percent: yield_to_maturity at par

5.28

percent: the file's 10 Yr

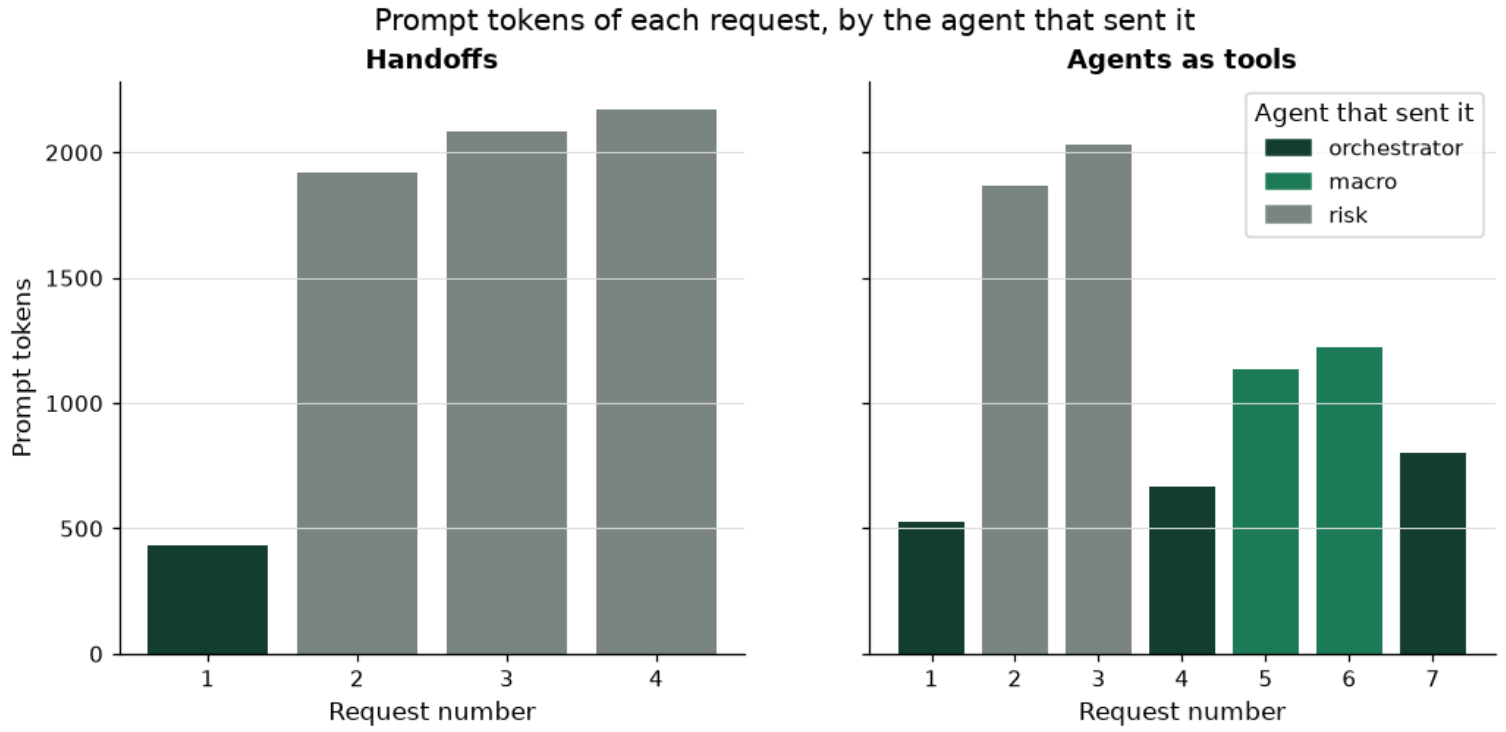
```
Total position DV01 in dollars of the synthetic holdings file: 306,996.493904 usd; pandas 306,996.493904; the...
10 Yr Treasury par yield on 2026-10-02: 4.0 pct; pandas 5.280000; the same at 2 decimals: no
```

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 4 requests, 6,610 prompt and 249 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

A 4 percent bond priced at 100 yields 4 percent, whatever the curve says. The tool was asked the wrong question.

One specialist received the whole question, and it needed two. The keys caught it; the checks could not.

Prompt tokens of each request



The risk specialist's requests are the largest in both runs: it carries eleven tools. The orchestrator carries none.

specialist and the allow-list, asserted

```
ValueError: risk's tools are ['dv01'], not the allow-list ['dv01_per_100']
```

```
macro           holds exactly its allow-list: 6 tools
credit analyst  holds exactly its allow-list: 3 tools
risk            holds exactly its allow-list: 11 tools
PM              holds exactly its allow-list: 0 tools
risk            offered in its saved requests: 11 tools, the allow-list exactly
macro           offered in its saved requests: 6 tools, the allow-list exactly
```

The orchestrator offered only handoffs and agent tools: it holds no data tool, so it fetches no number itself.

route_table: who handled which step

Handoffs: the specialist's steps are the run's

```
1 orchestrator handoff transfer_to_risk
2 orchestrator handoff result to risk
3 risk tool request portfolio_totals
4 risk tool result {"as_of": "2026-09-30", "bonds": 200, "dts_years_bp": 1064.114013, "duration_...
5 risk tool request yield_to_maturity
6 risk tool result {"tool_call_id": "call_ifpnXWdtxKgXkTYDvd1JEXut", "yield_pct": 4.0}
7 risk answer {"text":"As of 2026-09-30, the total position DV01 in dollars of the syntheti...
```

Agents as tools: the specialists run inside a tool

```
1 orchestrator tool request risk
2 orchestrator tool result {"figures": [{"name": "Total position DV01 in dollars", "tool_call_id"...
3 orchestrator tool request macro
4 orchestrator tool result {"figures": [{"name": "10 Yr Treasury par yield on 2026-10-02", "tool_...
5 orchestrator answer As of 2026-09-30, the total position DV01 in dollars of the synthetic holdi...
```

The route table says who handled each step; the request table, read from the exchanges, says who sent each request.

The desk brief: three specialists, then the PM

4	2	4	2	1
orchestrator	macro	credit analyst	risk	PM

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 13 requests, 17,559 prompt and 1,017 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

```
numbers in the brief that no tool result holds: []
the credit analyst quotes Section 4.12; the quote is in that section word for word: yes; the brief holds the ...
```

The failing check is the credit analyst's own: a figure 2.0 with unit to_1, under the key text.

The PM has no tool. Every number in the brief traces to the specialists' answers; it added nothing.

Parallel tool requests, once

```
parallel_tool_calls: desk False, parallel_desk True
```

2

tool requests in one reply

6

requests, against 7 off

4

replayed from the run before

`the log: orchestrator, risk, macro, risk, macro, orchestrator`

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 6 requests, 7,598 prompt and 368 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

Live, the interleaving depends on reply times: two runs can log two orders. So the course keeps it off.

What the orchestrator costs

```
run                agents   requests     prompt completion    largest        usd
handoffs                2          4       6610        249       2173   0.003042
agents as tools         3          7       8254        352       2030   0.003865
parallel on             3          6       7598        368       2030   0.003628
one agent               1          3       8066        115       2882   0.003410
desk brief              5         13      17559       1017       2185   0.008651
```

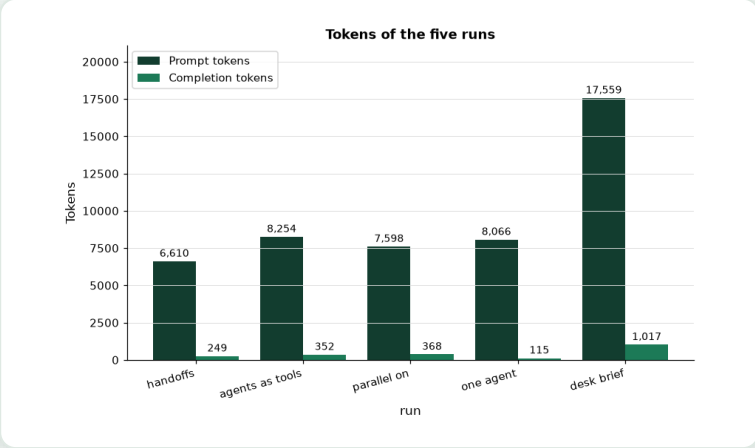
```
numbers in the answer that no tool result holds: ['298,998.53']
sector_totals returned 10 sectors, which add up to 306,996.49 dollars; pandas: 306,996.49
```

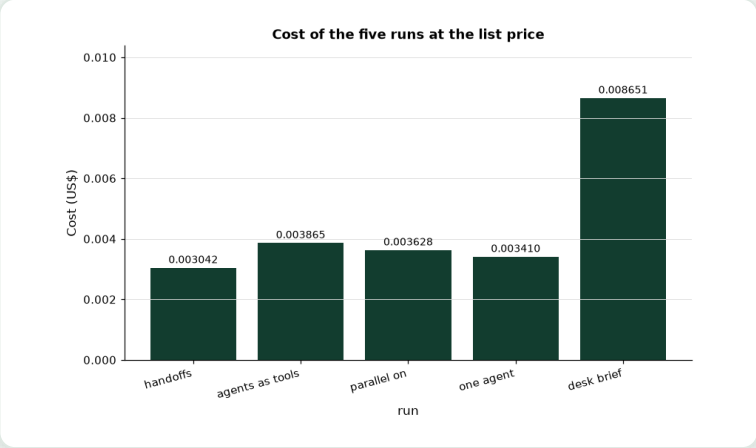
Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 3 requests, 8,066 prompt and 115 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

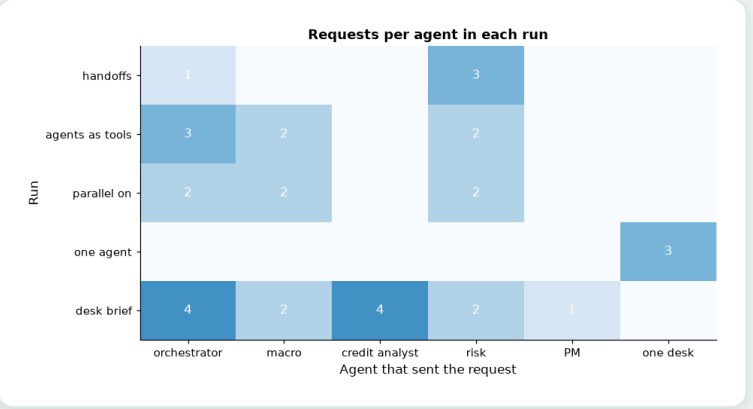
Twenty tools: the one agent took sector_totals and added the sectors itself, short of the key.

The split costs requests, and each request is smaller. Here it bought the right total.

The five runs in pictures







Run	Agents	Requests	Prompt	Completion	Largest	Cost (US$)
handoffs	2	4	6,610	249	2,173	0.003042
agents as tools	3	7	8,254	352	2,030	0.003865
parallel on	3	6	7,598	368	2,030	0.003628
one agent	1	3	8,066	115	2,882	0.003410
desk brief	5	13	17,559	1,017	2,185	0.008651

Excel: requests and tokens by agent with SUMIF

```
=SUMIF($B$2:$B$15,"orchestrator",D2:D15)      sum range E, F, G: tokens and dollars
```

orchestrator	10 requests, 7,588 prompt, 412 completion tokens
risk	9 requests, 17,880 prompt, 632 completion tokens
macro	6 requests, 7,112 prompt, 369 completion tokens
credit analyst	4 requests, 6,277 prompt, 353 completion tokens
PM	1 requests, 1,164 prompt, 220 completion tokens
one desk	3 requests, 8,066 prompt, 115 completion tokens
PM, typed pm	1 requests, 1,164 prompt, 220 completion tokens

```
pm              Excel 1 requests; pandas: no agent named 'pm'
```

Excel 16.0 build 20430, typed by script. SUMIF ignores case ("pm" finds PM); pandas' == does not.

Git: runs.csv by agent, and bring it home

```
agentkit day 11: specialist, route_table, 54 tests; runs.csv by agent
Start day 11: the earlier modules and agentkit as day 10 left it, 52 tests
agentkit.py      | 41 ++++++++++++++++++++++++++++++++++++++++++++
runs.csv         | 15 +++++++++++++++
test_agentkit.py | 24 ++++++++++++++++++++++++
3 files changed, 80 insertions(+)
```

```
python -m pytest -q test_agentkit.py          # 54 passed
git add agentkit.py test_agentkit.py
git commit -m "agentkit day 11: specialist, route_table"
```

One row per agent per run: git log -p runs.csv later shows which agent's numbers changed.

The AI check

The docstring is the prompt. The agent code that comes back runs and looks right.

1

Docstring

Exactly an allow-list: one job, its own tools, nothing else.

→

2

Code back

Written for the course in the style of an AI answer. It runs.

→

3

Test first

Build all four; assert each one's tools. The PM holds 20, dv01 among them.

→

4

Fix

The allow-list and nothing else, checked: 6, 3, 11, and 0 tools.

Today: specialist() hands every agent all 20 tools "so none gets stuck": the PM can compute a DV01 itself.

Code that runs is not the same as code that is right.

What you can do now

- Why a desk of agents splits the work
- specialist: tools equal to an allow-list, asserted
- Handoffs pass the conversation; tools keep control
- route_table and the request table, side by side
- A brief from an agent with no tools, checked

Week 3, Day 12 Memory and state